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Understanding Swap Spreads

November 2015

Swaps and bonds have the same type of market risk. They differ markedly in terms of liquidity and credit risk.

Swaps contain the credit risk embedded in the LIBOR index on the floating leg and, potentially, counterparty default risk.

Liquidity has separate drivers in the Swap and Treasury markets. Moreover, Treasuries perform a collection of unique functions in financial markets that give them a status very close to cash.

Swap Spreads

The difference between the yield of an interest rate swap and the yield on a Treasury of the same maturity

Components of a Swap Spread

- **Counterparty Default Risk:** Considered to be minimal because of credit risk-mitigation (mark-to-market, collateralization, termination options for long-dated swaps). However, as the Lehman bankruptcy demonstrated, re-establishing positions previously held with a now defaulted counterparty may be expensive, esp. if the counterparty is a significant market participant.
- **Liquidity Risk:** Treasuries are preferred instruments, Treasuries offer access to cheap repo (usually)
- **Credit Risk:** The credit spread in swap rates originates from the stream of LIBOR-indexed payments on the floating leg. LIBOR itself is a short-term and default-risk-embedded interbank lending rate. There is no idiosyncratic risk associated with a swap (unlike a bond) and swap spreads serve as a gauge of systemic risk.
- **Funding Risk:** Availability of repo, stability of repo haircuts, repo “specialness,” ability to rollover repo

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The theoretical calculation for swap spreads demonstrates that the fair value of the swap spread is a direct function of expectations of the future spread between 6-month LIBOR and 6-month repo.

Assumptions:

(1) It costs nothing to short a bond – in practice, bonds can go "special" in the repo market.

(2) There is no counterparty default risk in the swap (this allows us to use the risk-free yield to discount the fixed cash flows)

1 Stylized trade

- Short \$X of 10-year government bond at par and yielding Y
- Invest the proceeds in 6m GC repo and roll over at each 6m over 10 years
- Simultaneously enter a 10-year swap contract to receive fixed/pay 6m LIBOR at current swap rate R on an \$X mm notional principal

2 Semi-annual cash-flows for trade (over 10 year life of the bond)

$$\left[\frac{R-Y}{2} - \frac{LIB_6 - GC_6}{2}\right] \times X$$

3 Present value at initiation

$$\sum_{i=1}^{20} \frac{(R-Y)X/2}{\left(1+\frac{r_i}{2}\right)^i} - \mathbb{E}\sum_{i=1}^{20} \frac{(LIB_6 - GC_6)X/2}{\left(1+\frac{y_i}{2}\right)^i}$$

$$y_i = r_i + \gamma$$

r_i: riskless nominal zero-coupon yield

γ: risk premium for future movements in the LIBOR-Repo spread

4 Swap Spread

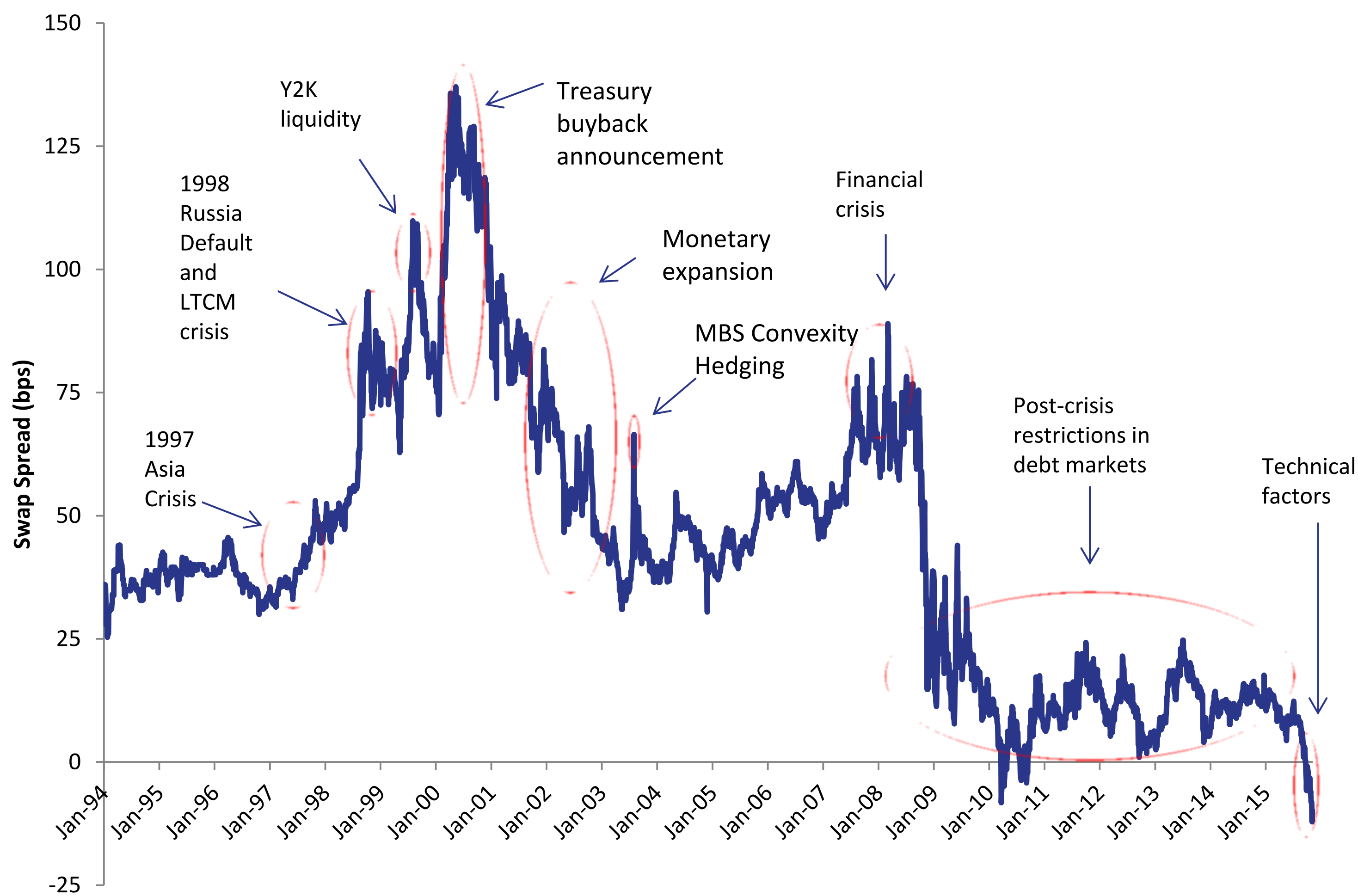
- If both the government bond and swap markets are fairly valued

$$\text{Swap Spread} = R - Y = \frac{\mathbb{E}\sum_{i=1}^{20}\left[(LIB_6 - GC_6)/\left(1+\frac{y_i}{2}\right)^i\right]}{\sum_{i=1}^{20}\left[1/\left(1+\frac{r_i}{2}\right)^i\right]}$$

- If risk premium γ = 0, then the fair swap spread is the weighted average of the future expectation of this spread.

1994-1997: Swap-spread levels and volatilities were relatively low.
July 1997: Asian crisis leads to credit risk concerns
August 1998: Russian crisis and the near collapse of LTCM spark flight-to-quality
Mid-1999-2000: Swaps spreads were kept high by concerns about potential Y2K disruptions.
2000: Swap spreads widen because of large UST buybacks
2001-2003: Fed rate cuts and fiscal surpluses giving way to deficits
Early August 2003: MBS Convexity Hedging
2007-2009: Financial crisis leads to wider swap spreads with flight-to-quality
2009-2015: Deficit of risk (equity) capital + greater focus on (repo) counterparty risk + higher haircuts + higher repo rates
August 2015-Present: Foreign selling of UST, bloated dealer balance-sheets, increasing debt issuance by FIs, reevaluation of duration hedges, stop-outs

HISTORY OF THE 10-YEAR SWAP SPREAD
(January 1994 – November 2015)



Source: Citigroup

The following factors all show some correlation with swap spreads. They can be decomposed into **fundamental factors** (Carry/funding effect, Systemic risk) and **technical factors** (Treasury-specific factors, supply/demand effects).

None of these factors necessarily account for the impact of new regulatory frameworks such as Dodd-Frank, Basel 3 RWA, LCR, Supplementary Leverage Ratio, Mandatory Swaps Clearing etc.

Factor	Comments
Treasury Supply	Dynamics of fiscal deficit and government debt
AA Credit Spreads	Changing perceptions of credit risk over the long term, particularly in the AA-rated sector, as the majority of banks quoting LIBOR rates are AA-rated
On-Off-the-Run Treasury Bond Yield Differential	The difference in yield between on-the-run and off-the-run Treasuries constitutes an indication of the "liquidity convenience yield"
Slope of the Yield Curve	Swap spreads tighten as the yield curve gets steeper and vice-versa: in steep yield curve environments the cost of funding long-dated fixed-rate liabilities increases
LIBOR-Repo Spread	Captures the LIBOR credit spread and the liquidity premium stemming from the ability to fund Treasuries in the repo market
MBS Duration	Swaps spreads tighten as mortgage rates decrease (duration buybacks) and vice-versa
Volatility of the Stock Index	Stock market volatility is good indicator of the overall risk aversion in the capital markets

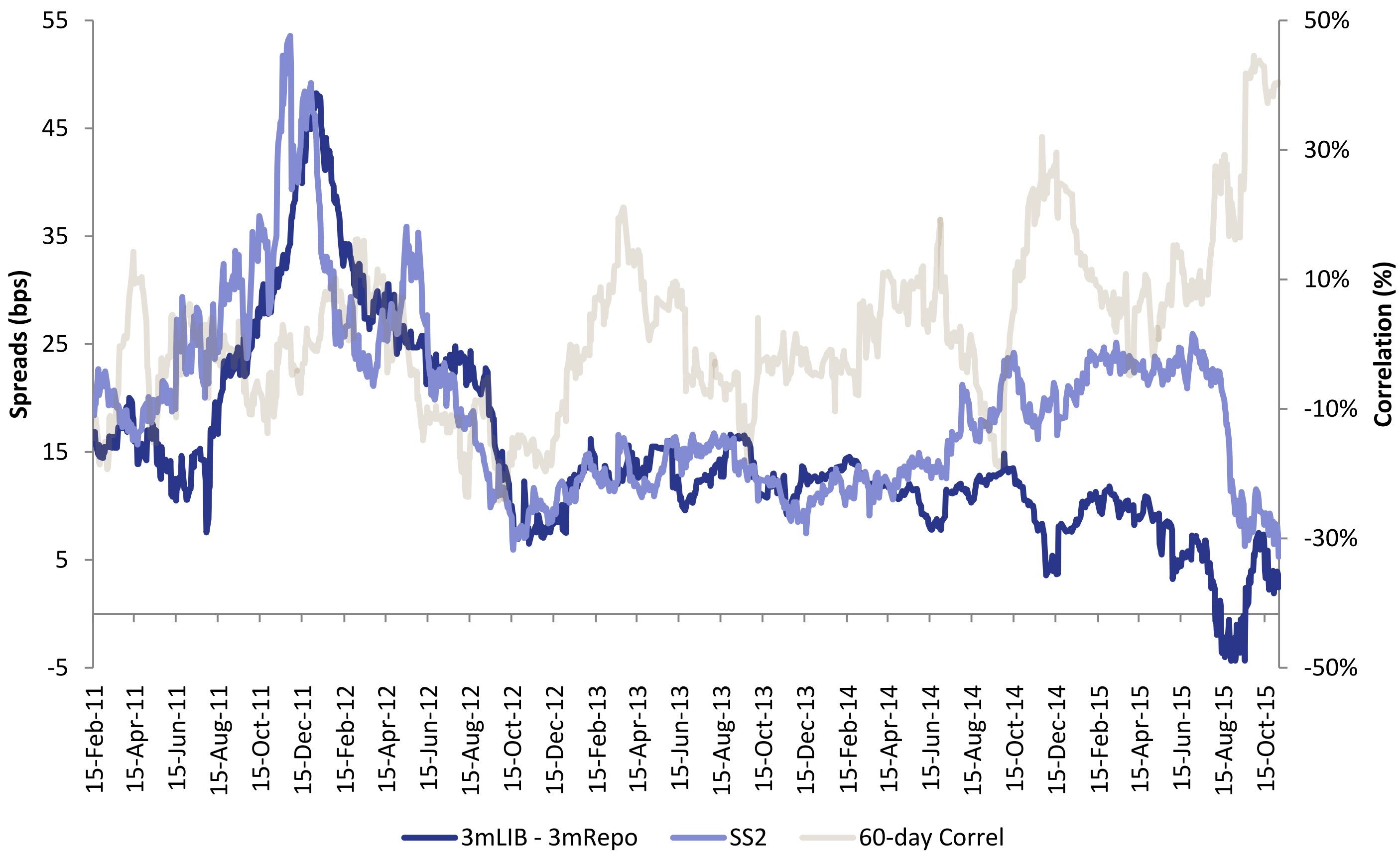
Theory argues that swap spreads should reflect the expected values of the LIBOR-Repo spread over the term of the swap. The theory should hold more weight for short-end spreads where the holding periods are shorter.

In practice, several technical factors affect the swap spread and have a more pronounced effect in one regime compared to another.

For example, 2-year swap spreads widened materially relative to the LIBOR-Repo spread in the middle of 2014 because of various technical reasons including lower short-maturity Treasury issuance, money market reform, and expectations of a wider reverse repo program/interest on excess reserves band.

More recently, the swap spread has plunged due to balance sheet pressures faced by FIs.

THE 2-YEAR SWAP SPREAD vs LIBOR-REPO Spread
(February 2011 – November 2015)



Source: Morgan Stanley, Cello

There are plenty of natural and very large receivers in the swap market. There is no natural payer that matches the receivers in size.

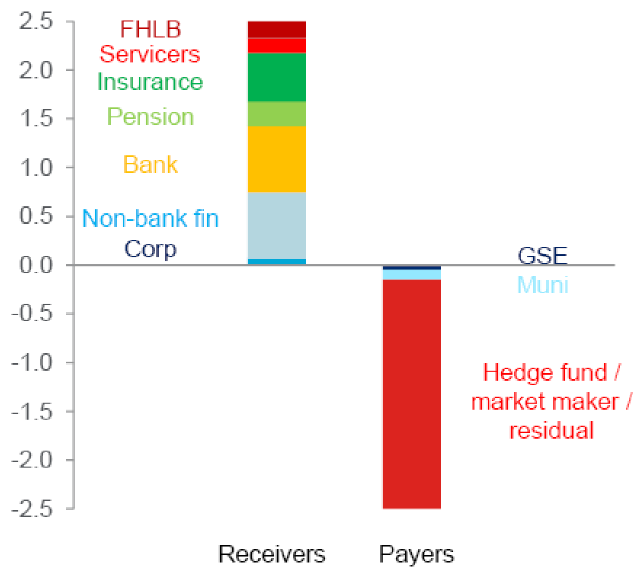
Accounts that need short term duration hedges against other cash products have turned to listed Treasury Futures instead of Swaps because of lower initial margin costs and swap spread behavior.

CHANGES IN SWAP CLEARING

Swaps clearing

A market out of balance…

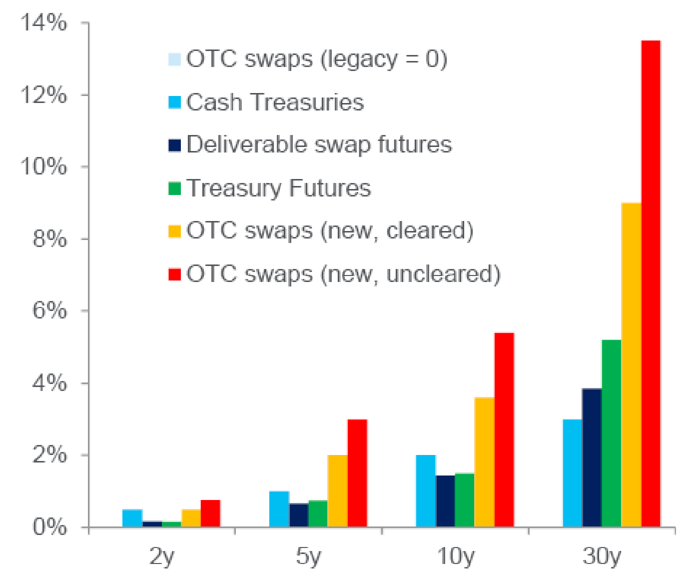
Imbalance between OTC swaps payers and receivers, $bn DV01



Source: Dealer estimates.

…even before margins were hiked

Initial margin requirements (% notional)*



Source: CFTC. * Calculated from current VaR levels.

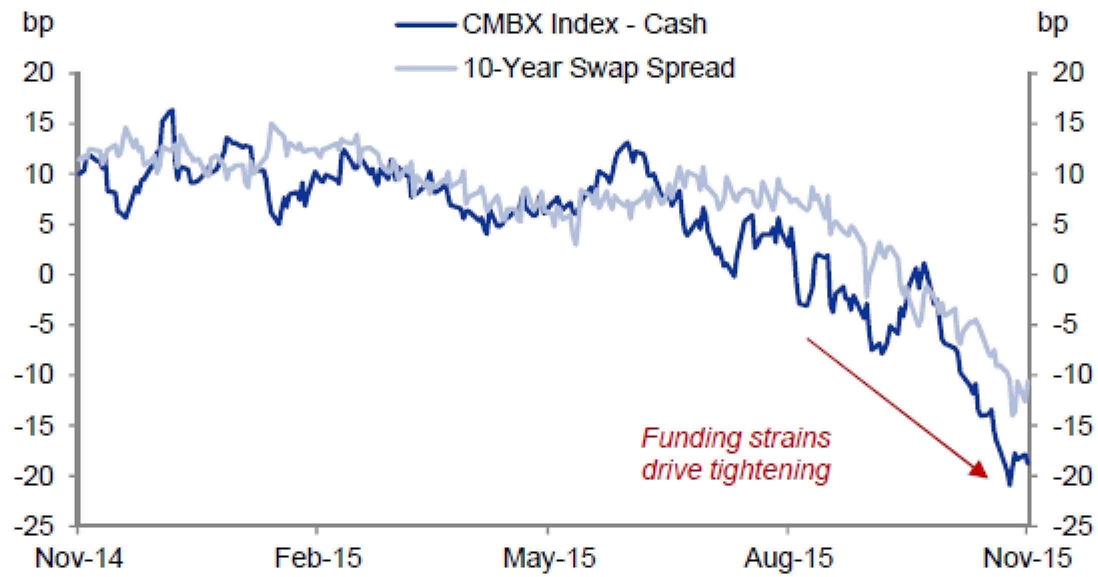
Activity migrating from swaps towards futures

Source: US Treasury (July 2013)

The recent swap spread tightening can also be seen in other asset classes such as CMBS or corporate credit. The general underperformance of cash products versus synthetic products is indicative of how funding pressures are an important driver of recent swap spread tightening.

SYNTHETIC vs CASH BASIS: CMBS
(November 2014 – November 2015)

Exhibit 2: The CMBX index vs. CMBS cash bond basis has tightened as swap spreads have tightened
10-year swap spread and CMBX7 AAA vs. CMBS cash bond basis



Source: IDC, Markit and Goldman Sachs Global Investment Research

Source: Goldman Sachs

A swap spread arbitrage trade may need to be held until maturity consuming balance sheet and relies fundamentally on the smooth functioning of repo markets.

Specifically, swap spread arbitrage requires (1) stable levels of risk capital, (2) the ability to obtain repo (or some other source of financing), and (3) stable repo haircuts and financing rates (for financing the Treasury side of the trade).

A negative swap spread may thus be fair compensation for assuming repo refinancing risk over a long term.

There is no intrinsic contradiction in GC Repo (collateralized) trading above LIBOR (uncollateralized) – they reflect rates applicable to different sets of borrowers.

30-Year Treasury Purchase

- Receive 3.02% p.a. for the next 30 years
- Pay 1-month repo, rolled every 1 month, to finance the purchase

30-Year Swap Trade

- Pay 2.53% p.a. for next 30 years
- Receive 1-month LIBOR rate, reset every 1 month, for next 30 years

Net Cash Flow: Long 30-Year Treasury & Pay 30-Year Fixed

- Fixed-rate differential = Treasury Rate – Swap Rate ≈ 49 bps
- Floating-rate differential = LIBOR – Repo ≈ -5bps

Risks

- **Duration**: Limited to none
- **Mark-to-market**: Swap spreads tighten/widen -> MTM Loss/Gain
- **LIBOR-Repo Spread**: Spreads widen/tighten -> Carry increases/decreases
- **Balance Sheet**: Haircuts increase/decrease -> Trade ROE decreases/increases; Liquidity decreases/increases

Source: Citigroup, Cello. As of 11/20/2015.

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